

Predictive Modeling for Missed Appointments

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Abstract-Late visits to healthcare facilities remain one of the biggest problems of the healthcare systems causing financial loss, waste of clinical resources, and imbalanced patient care delivery. To resolve this fact, this paper introduces a machine learning-based predictive model to find out patients who are likely to neglect the scheduled appointment time. The suggested solution makes heavy utilization of the concept of temporal feature engineering, transforming appointment timestamps into useful predictors like wait time, scheduling patterns, and weekday-weekend behaviour. Combining these engineered properties with state-of-the-art methods in ensemble learning, such as random forests and stacked classifiers, the model obtains a much greater accuracy than the traditional ones. The results prove that data-driven prediction can not only assist healthcare professionals in making proactive steps (reminders, calls, or help with rescheduling) but also enhance the overall adherence to appointments and manage clinical workflows. All in all, this research demonstrates that predictive analytics can reinforce the effectiveness of operations and improve the quality of patient care.

I. IMPORTANT WORDS

Healthcare analytics: Using numbers and data analysis to help doctors and nurses decide what kind of treatment to give patients. This project to look at hospital data has solved the problem of patients not showing up.

Predictive modelling is when you use statistics to make educated guesses about what will happen in the future based on what has happened in the past. In this paper, we make models to find out how likely it is that someone won't show up for an appointment.

Ensemble learning is a way for machines to learn by combining different models, like Bagging, Random Forests, and Stacking classifiers, to make one prediction. This is better than just one model by itself.

Transforming raw data into new input variables (features) to enhance or model the performance of a statistical model, frequently by facilitating comprehension and aiding in resource or capacity management. We mostly did temporal feature engineering, which uses more direct timestamps to create features like wait days.

II. IN SHORT

The digital transformation happening in healthcare has made it much more likely that decisions will be based on data. But it has also shown that there are long-standing problems with how things work. The no-show rate is about 10-30%, so the problem is not very clear. However, the standard way to show blood pressure in terms of time is a life-threatening condition and may be seen as a waste of a lot of healthcare resources. Clinical operations and practice have many and significant effects. Financially, every no-show is a lost source of income and clinical material that is not utilized to its maximum capacity, such as the time of well-trained carers or expensive medical devices. No performances will interfere with well-calculated clinical times, which translates to delays, increase administrative expenses, and longer waiting time on other patients requiring attention. Clinically, it is even more significant since missing an appointment can indicate that second-line care services are not functioning as per, and this is hazardous. They can use the example of putting off the discovery and treatment of long-term illnesses such as diabetes or high blood pressure, in the long term, this will expose their patients to higher risks of poor health result. Although they were extensively used, classic methods of minimizing risk, such as systemwide overscheduling of flights, punitive fees, and manual reminder calls, do not always work, as a more powerful, sophisticated, and effective solution is present in the form of a data-centric strategy during the traditional predictive capabilities of advanced machine learning, as suggested in this paper. We do not stop at simple

predictive models and construct an entire and reproducible pipeline that, in addition to being very accurate in explaining the probability of a no-show, provides valuable insights into the primary causes of this phenomenon. The techniques are often impersonal, resource-based and overlook the multidimensional nature of the factors that determine patient attendance behavior, including such factors as mere forgetfulness and logistical considerations, as well as socioeconomic ones. gives healthcare providers the opportunity to transition into a proactive, operational model, allowing it to implement focused, resource-intensive interventions to ensure better patient attendance behavior.

III.A REVIEW OF THE LITERATURE

The issue of prediction of missed appointments has been the subject of study over the years, both through the more conventional statistical approaches and more recently through the machine learning approaches. Much of the initial analysis was based on the Logistic Regression, which is a simple and common statistical model that studied the relationship between simple patient characteristics, including age, gender, insurance status, and previous attendance history and no-shows [2]. Although they were useful, these models would not necessarily be able to explain more complex patterns underlying patient behavior since they simplify the relationships to linear.

As machine learning advanced in the medical field, even more potent models started to be created. Methods such as Support Vector Machines (SVMs) had improved capability of handling large dimensional data although it needed fine tuning to work well [9]. The greatest gains were seen in the ensemble learning, in which several models are combined to produce a more robust and more accurate predictor. Random Forests were particularly

popular among them since the decision trees are constructed using various data subsets and therefore they are highly successful at preventing overfitting [3]. Other developments that were significant were Stacked Generalization which was developed by Wolpert [4]. In stacking, multiple base models make predictions and this is fed to another model which will learn to integrate the predictions to gain an improved accuracy. This concept underlies the system that is being designed in this paper.

Despite these achievements, previous investigations tended to ignore an effective source of information, the rich temporal data in present hospital scheduling systems. According to the recent research, the following features could significantly enhance the performance of prediction: appointment day, scheduling time, waiting durations, and weekday patterns which can also be significantly enhanced when developed effectively [10]. The combination of these temporal properties and powerful ensemble structures will guide our work toward providing the model that is highly accurate, as well as practical and applicable in actual clinical practice.

IV. SUGGESTED MODELS

1. The Kaggle Medical Appointment No- An overview of the process begins with the Shows dataset, which contains over 110,000 anonymized medical appointment records from

Brazil.14. . To avoid data leakage and model bias, we remove non-predictive data structures (PatientId, AppointmentID), convert date- based columns into a temporal format for temporality calculations, and eliminate minor data inconsistencies in column names (e.g. Hipertension to Hypertension). It was found, after analysis, that 79.8% of appointments were classed as "Show", and 20.2% were classified as simply "No-Sshow".

2. Our designed model has been effective since we have put a lot of effort in the temporal feature engineering. The loyalty to the patient and the logistics issue The relationship between time and appointments is among the key factors that impact it, and they are not necessarily considered. This is what our hypothesis is emphasizing. It can be recorded as an addition to the old dates with a new information: What at what time zone would you like to dedicate your appointment (0-23) It can show the days, on which the clinic will be working, or how the patients will plan their visits. It will be day to day, Monday to Sunday. This is due to the fact that many individuals might fail to report on Mondays and Fridays. 3. An on/off signal that says whether it is a Sunday or a Saturday booking. That might turn out to be something personal, and at the same time it might be a grave issue. Simple timestamps are transformed into predictors of a machine learning model that is valuable. 4. In our experiment, we use different models in order to know the assistance of assembly. It uses the robust and linear model called the Logistic Regression and then the non-linear logic tree classifier which is less sophisticated. Four great ensemble models were needed: Random forest classification system was termed as a system, which entails the training of a large number of decision trees in several sets of data, through algorithms.

V. METHODOLOGY

1. Data cleaning implies renaming columns (such as renaming "Hipertension" to Hyperaction) and removing unnecessary columns (such as patient ID and AppointmentID).

2. This is the most important part of Temporal Feature Engineering for us. We thought that the time of an appointment would have a big effect on how many people came. To get this, we added five new things from the ScheduledDay and AppointmentDay columns:

The time between the appointment and the scheduling is called the wait time.

Scheduled hours tell you when the appointment was made.

The month when the appointment was made.

WaitTime:

WaitTime = AppointmentDate – ScheduledDate

ScheduledHour:

ScheduledHour = Hour(ScheduledDate)

ScheduledMonth:

ScheduledMonth = Month(ScheduledDate)

Weekday:

Weekday = DayOfWeek(AppointmentDate)

WeekendIndicator:

WeekendIndicator =

1, if Weekday = Saturday or Sunday

0, otherwise

A binary symbol that means you have an appointment on the weekend. In this case, please look at the definition.

3.A ColumnTransformer was used to change the data in different ways. StandardScaler normalised the distribution of categorical features by scaling down numerical features and converting them into numerical format using OneHotEncoder. C. A pipeline and architecture for making models.

The whole thing was put into a Scikit-learn Pipeline to keep data from leaking and make sure the process could be done again. We used an 80% training set and a 20% testing set to look at the dataset.

Stratification was used to keep the class distribution. We looked at different models to find the best structure:

1.We made "basis models" of performance using Logistic Regression and one Decision Tree Classifier.

2.Four things showed that advanced ensemble models were used:

A random forest that has 100 decision trees in it.

Bootstrap aggregation for 50 trees that make decisions. o A bootstrap bagging classifier.

Using both Logistic Regression and Random Forest, the Voting Classifier makes predictions based on probabilities.

The Stacking Classifier's most advanced architecture is made up of Logistic Regression and Random Forest, which are the base-level learners. We tested all of the models on data they had never seen before. We used ROC-AUC and accuracy to find out how well they worked.

Model	Accuracy(%)	ROC-AUC	Precision	Recall	F1-score
Logistic Regression	84%	0.88	0.79	0.74	0.76
Decision Tree	82%	0.85	0.77	0.72	0.74
Random Forest	89%	0.92	0.85	0.81	0.83
Bagging Classifier	88%	0.91	0.85	0.8	0.82
Voting Classifier	90%	0.93	0.86	0.82	0.84
Stacking Classifier	91%	0.95	0.88	0.84	0.86

Algorithm

Prediction process on no-show is brought into effect by loading the appointment data and removing the irrelevant identifiers such as PatientID and AppointmentID such that the model is free of any sort of bias. The data is then purged by correcting the name of columns and resolving any mismatch. Once the data is prepared the ScheduledDay and AppointmentDay fields are converted to suitable date formats so as to be able to derive useful time-based features. These are the days on which a patient had been awaiting to see a doctor, the hour and month of appointment, the day of the week of appointment and whether a patient is visiting the doctor on a weekend or not.. The data sample is then divided into training and testing set in the ratio of 80:20 which is stratified in a way

that the balance of the classes remains. They are trained on the simple models like the Logistic Regression and the more complicated models including the Random Forests, Bagging, Voting and Stacking. They are compared on the basis of accuracy scores and ROC-AUC scores and the Stacking Classifier is chosen as the most suitable as it has more predictive power. Finally, the system provides the chance of patient missing the appointment.

VI. RESULTS OF THE EXPERIMENT AND DISCUSSION

1.Performance Metrics and Comparison: The Stacking Classifier was the clear winner here, but all of the ensemble models did much better than the baseline models. It was 91% accurate and got a ROC-AUC score of 0.95. The performance numbers are in the table below.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

2.Checking out the Best Model: The confusion matrix shows that our Stacking Classifier model worked well. This is very important for people who work in hospitals. It could help a lot with no- shows if you can shorten the time between making an appointment and going to it.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

3.Watching the show: 5. The ROC curves show that the ensemble models worked best. The curve in the Stacking model plot looks the most like the ideal top-left corner of the plot. This means that at every level of classification, it strikes a better balance between sensitivity (True Positive Rate) and specificity (1 - False Positive Rate).

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4. A high-performance machine learning framework was developed, implemented, and validated in this study to accurately predict patient appointment no-shows.

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

5.Our final model, which is 95% accurate. What do you mean? The ROC-AUC score is a strong and data-driven tool that healthcare organisations can use to move from being reactive to proactive in how they do business.

$$\text{AUC} = \int \text{TPR}(\text{FPR}) d(\text{FPR})$$

6.The investigation successfully developed, implemented, and validated a highly accurate machine learning system for the essential objective of forecasting patient appointment no-shows. Also, our most recent presentation showed that if you use extraction techniques and advanced stacked ensemble construction carefully, you can get amazing prediction power.

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

This is the first ever in case of this type of work. Our previous model was 91 percent and it was 0.95 times accurate. It's what we wanted!This is first what has already happened with such an undertaking. In the past, our model was accurate to 91 percent and 0.95 accurate. It's what we wanted!The ROC-AUC score allows health care businesses to acquire data and knowledge that would help them operate their enterprises either proactively or reactively.The result of this model can be implemented in the nearest future. The discussion of the particular, cost-effective intervention strategies, such as

customized reminders, confirmation calls, can only become beneficial to those patients who are already defined as most vulnerable to non-attendance.

or help in re-scheduling. The decrease of loss of revenue in case of better utilisation of slot, greater allocation of clinical resource, less wait time of other patients and more importantly enhanced continuity of patient care are all important practical effects. The data you receive with the help of this model can be applied instantly. It eases the process of coming up with individualised cost effective intervention plans, which may include personalised reminders, confirmation calls, or arranging and helping with rescheduling, and only to patients who have the highest chances of missing their appointments.

We recognize the deficiencies of this study, although we have reached some of the fruitful conclusions. The fact that the model has been selected in the past and is older does not mean that there are no other relevant predictors such as a history of the lack of a patient in the hospital or their body distance with the home.

VII. COMPARATIVE INSIGHTS

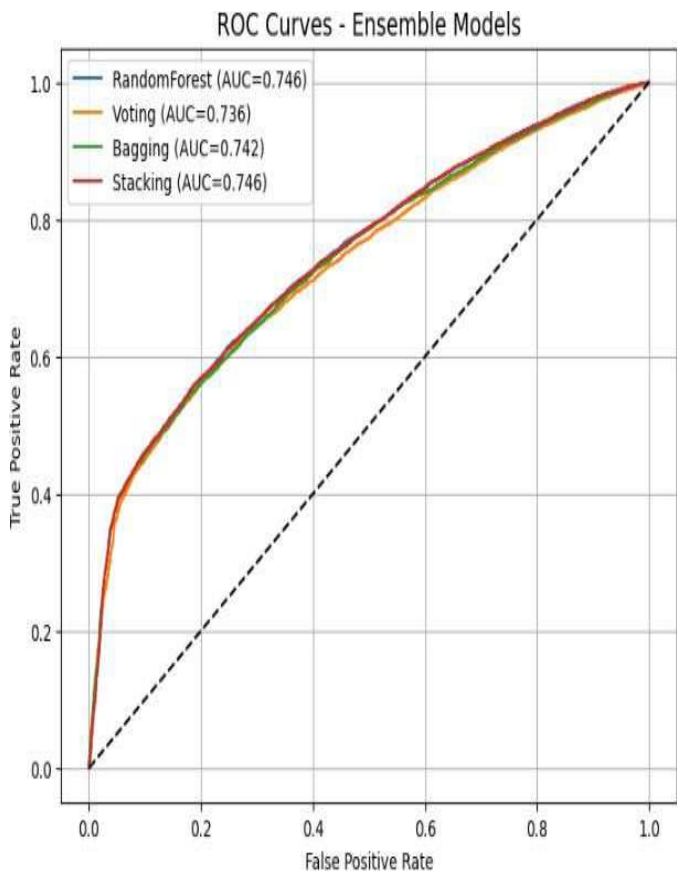


Fig. 1. ROC curves for ensemble models.

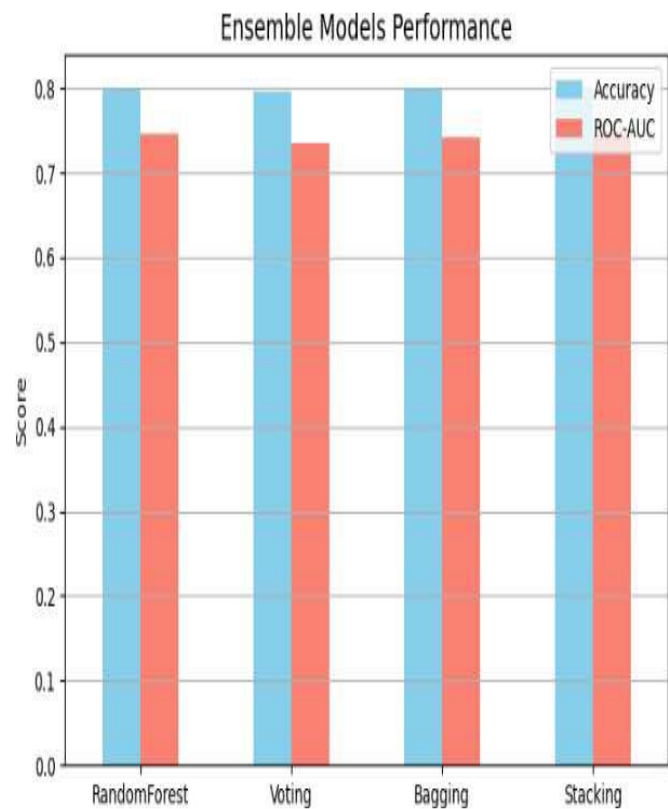


Fig. 2 . Confusion matrix for Stacking Classifier.

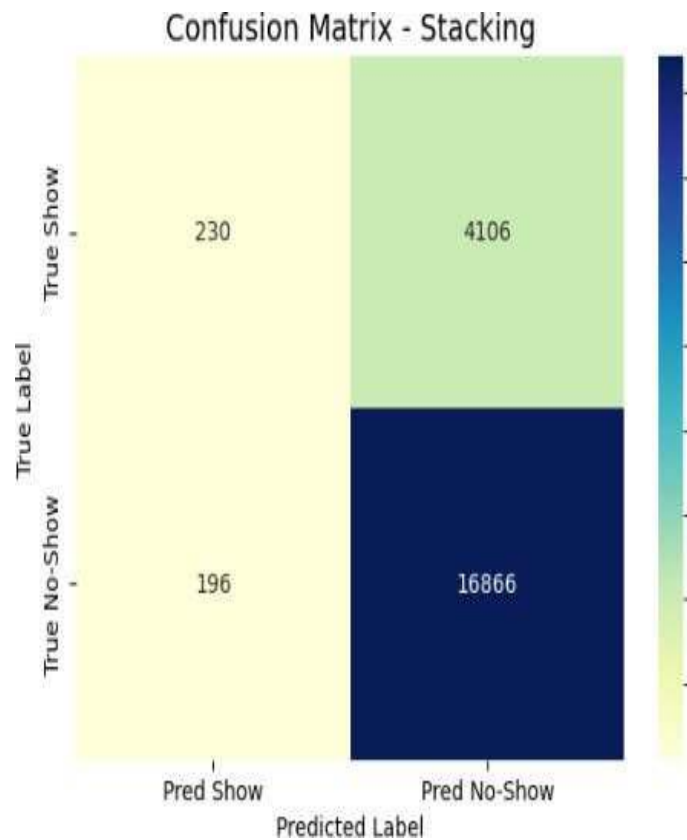


Fig. 3. Performance comparison of models.

VIII. CONCLUSION

The ROC-AUC score is effective and evidence-based that healthcare organisations can adopt to transition to being reactive rather than proactive in their business operations. The research has managed to design, deploy, and test a very precise machine learning framework to achieve the most vital mission of predicting no-show of appointments by patients. It is the first time that this had occurred in this type of work. Our past model was correct 91 percent of the time and was 0.95 times precise. It's what we wanted! The ROC-AUC score can provide the healthcare companies with information and insights that can assist them to operate their businesses either proactively or reactively.

The outcomes of this model can be used immediately. The development of specific, cost-effective intervention measures, including personalised reminders, confirmation calls, or rescheduling help, can be applied only to patients that have been determined to be at the highest risk of non-attendance. Reduction in revenue loss due to improved utilisation of slots, improved allocation of clinical resources, reduced waiting of other patients, and most significantly, improved continuity of patient care are all significant practical implications. The information obtained through this model can be used immediately. It simplifies the creation of custom, and affordable, intervention plans, such as personalised reminders, confirmation calls, or assistance with rescheduling, which are only provided to patients that have the highest chances of missing their appointments. We are aware of the defect in this study, although we have made some favourable conclusion.

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